Department of Computer Science
National University of Computer and Emerging Sciences,
Multan Campus

CL2001 - Data Structures Lab

Lab 04 - Fall 2026

Marks Detail

Question/Task No	Q1	Q2	Q3	Q4	Analysis	Total
LLO Number	1	1	1	1	4	
Total Marks	6	6	10	14	4	**40**

Rubrics

CL2001 – DATA STRUCTURES LAB						
LLO #	**Statement ↓ Score →**	**Exemplary (100%)**	**Proficient (75%)**	**Developing (50%)**	**Beginning (25%)**	**Novic (1%)**
2	To **implement** and traverse doubly linked lists and circular linked lists.	Correctly implements all concepts, with working solution and clean/commented code and meet all lab requirements.	Implements all concepts correctly with minor logic bugs and solution working.	Implements all concepts but with major errors and solution not working	Solution is incorrect and all requirements of lab are not satisfied	Does not Work
4	To **behave** responsibly as an individual and in a team to ensure professionalism in lab experiments and project.	Takes ownership and works actively throughout the lab	Actively works and participates in-group discussions.	Partially work and participate in group discussions	Needs help from others group to complete assigned task	Does not Work

1. **Problem Analysis**: Start by thoroughly understanding the problem statement. Then, draft your logic on paper before proceeding to code.
2. **Coding**: After completing your pencil-and-paperwork, implement the solution in C++ using Microsoft Visual Studio.
3. **Code Quality**: Use descriptive names for variables and ensure that your code is properly indented for readability
4. **Comments**: Include a block comment at the beginning of each code segment to describe its functionality. Additionally, provide comments for each function, as well as for variables, loops, and classes, to enhance code clarity and understanding.
5. **Documentation**: Create a Microsoft Word document that includes: all your C++ code and the screenshots of the output for each task.
6. **Submission**: attach the C++ files to the Word document and submit the Word document and the .cpp files in a single archive (i.e. Lab0x_Problem_0x.cpp) via Google Classroom or Moodle.
7. **File Naming**: Use the format 23F1234_Lab0x for your files.
8. **Deadline**: Submit your assignment by the deadline. Late submissions or email submissions will not be accepted.
9. **Originality**: Ensure that all code is original. Copying from external sources and using generative AI will result in penalties.

The task-specific rubric should also be followed to ensure academic compliance.

Topics

- Pointers
- Pointers to Functions
- Link list (Single, Double, Circular)

Problem: 1 | 1+2+2+1

Write a program to determine whether a given singly linked list is a palindrome or not. A linked list is considered a palindrome if its elements appear in the same order when read from left to right and right to left.

Example



Note: All operations must be performed on one link list.

Problem: 2 | 1+2+2+1

Write a program to delete all occurrences of a given value from a singly linked list. The program should traverse the linked list and remove every node whose data matches the given value.

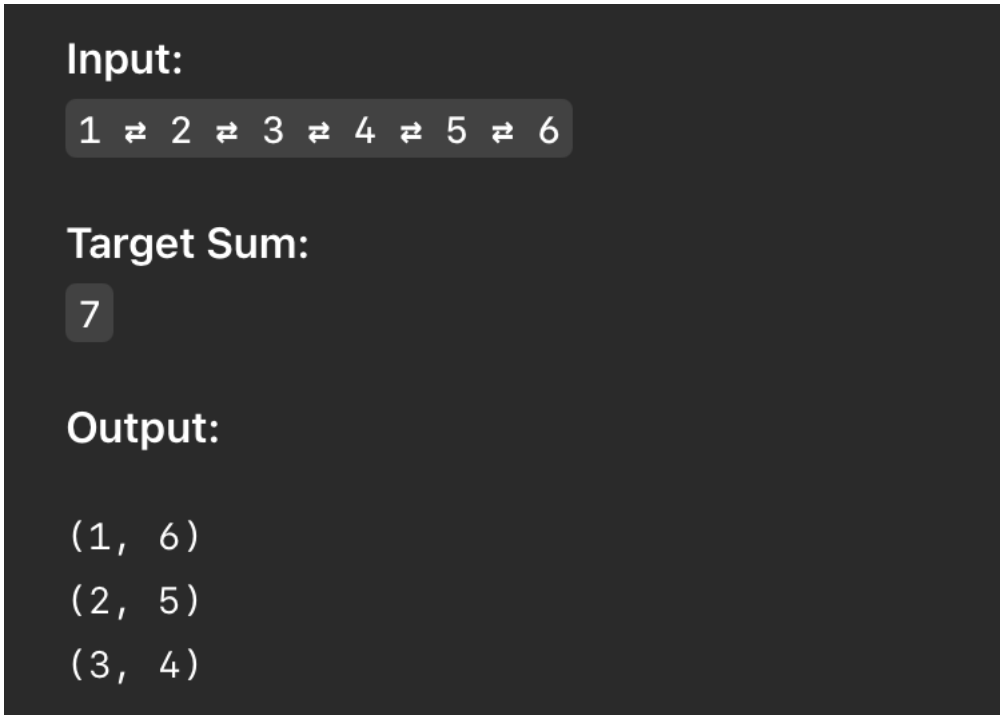
Example



Problem: 3 | 1+2+6+1

Write a C++ program to find and display all pairs of nodes in a doubly linked list whose values add up to a given target sum. Given a doubly linked list and a target value, identify every **unique pair (a, b)** such that:

a + b = target



Note: Link list and target will by dynamic and user defined.

Problem: 4 | 2+8+2+2

In the game of Musical Chairs, N players sit in a circle. Music starts playing, and the players move around the circle. When the music stops, the player at a particular position is eliminated because there is one fewer chair available. For this problem, assume that the music stops after every K positions. Starting from the first player, count K players in a circular manner. The player at the K-th position is eliminated. After eliminating that player, counting continues from the next remaining player. The process continues until only one player remains. The remaining player is the winner.

Write a C++ program using a circular linked list to simulate the Musical Chairs game.

Example

For:

N = 6

K = 3

Initially:

1 → 2 → 3 → 4 → 5 → 6 → back to 1

The players are eliminated one by one according to the counting rule.

Your program should display:

Eliminated: 3

Eliminated: 6

Eliminated: 4

Eliminated: 2

Eliminated: 5

Winner: 1

Hint: Remember that a circular linked list does not end at NULL.

Instead: last node → first node

Therefore, when traversing the list, you need to recognize when you have come back to the starting point.

The main challenge is to correctly handle deletion and continue counting from the next player without breaking the circular structure.

Note: To get full marks in this question, implement it using single circular link list.

```
Initial:
1 → 2 → 3 → 4 → 5 → 6 → back to 1

Count 1, 2, 3
Eliminate: 3

Remaining:
1 → 2 → 4 → 5 → 6 → back to 1


Count 4, 5, 6
Eliminate: 6

Remaining:
1 → 2 → 4 → 5 → back to 1


Count 1, 2, 4
Eliminate: 4

Remaining:
1 → 2 → 5 → back to 1


Count 5, 1, 2
Eliminate: 2
```

```
Remaining:
1 → 5 → back to 1


Count 5, 1, 5
Eliminate: 5

Remaining:
1
```